

JKBOSE

2026 CLASS 11TH

VECTOR

LECTURE-3

PHYSICS

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Topics To Be Covered

- 1** Revision ✓
- 2** Multiplication of two vector ✓
- 3** Cross product ✓
- 4** Feel of cross product ✓
- 5** Dot Product ✓
- 6** Resolution of Vector ✓

*“Your Attitude Shapes
Your Reality”*

- The $\vec{p}$ have magnitude and direction



$\vec{A}$ ✓

(i) Position vector



(ii) Displacement V



(vi) unit vector $\hat{a}$
 $m=1$
 Direction

(iii) Parallel vector

→ $\vec{A} \parallel \vec{B}$ $\theta = 0$

(iii) Antiparallel Vector



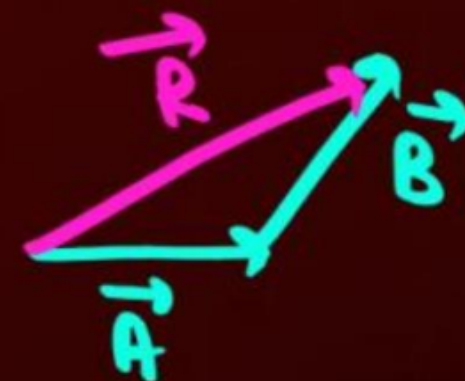
(iv) Equal Vector

→ $\vec{A} = \vec{A}$

(v) orthogonal vector



Triangular law of vector Addition



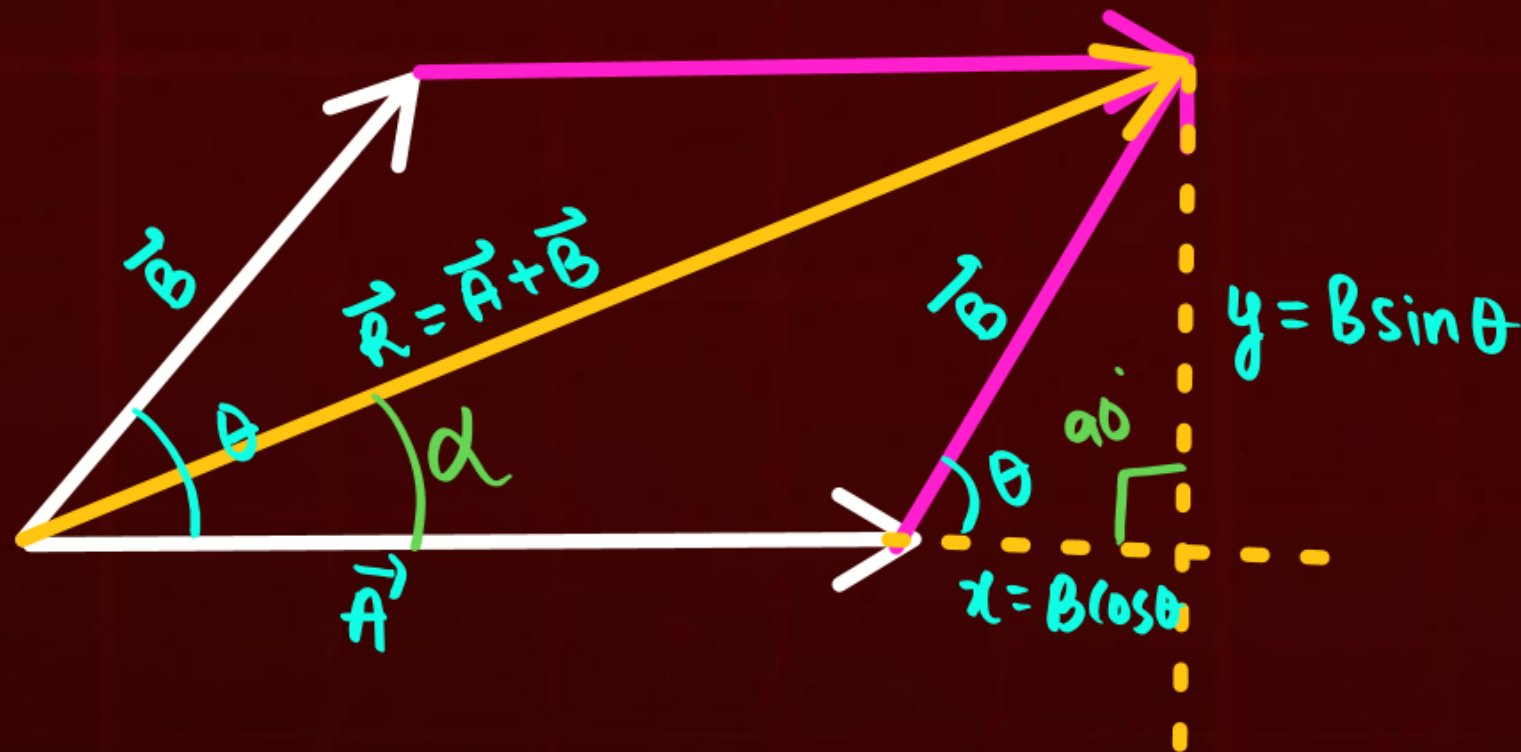
vector
algebra
+ 1 x ÷

Recap from 2nd Lecture

Parallelogram law of vector Addition

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$



lecture no. (3)

(maths) $\rightarrow$ Algebra.

(i) Addition of two vector

$$\vec{R} = \vec{A} + \vec{B}$$

Δ law of VA

119m law of VA

5 mark

x(ii) Subtraction of two
vector

$$\vec{R} = \vec{A} - \vec{B} \quad \times$$

(iii) Multiplication of two Vector. (X)

x(iv) Division $\div$

Multiplication of two vector

1. Multiplication with integer (number)

(a) multiply with positive integer

$$\vec{A} \times 2 = 2\vec{A}$$

magnitude ↑ Direction ×

$$\longrightarrow \times 2 = \longrightarrow$$

(b) multiplication with negative integer

$$\vec{A} \times -2 = -2\vec{A}$$

magnitude ↑↑ Direction change

$$\longrightarrow \times -2 = \longleftarrow$$

Multiplication of two vector

2. Vector Multiply with vector

$\vec{A}$ multiply $\vec{B}$

Rotation

Part - ①

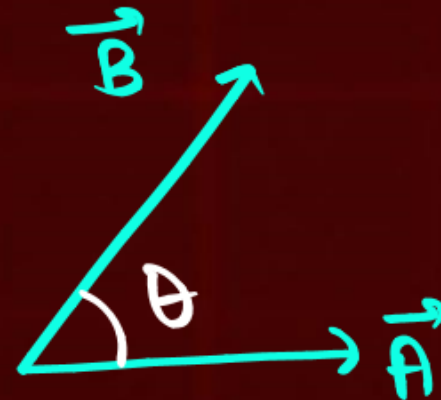
Part - ②

① Cross product
Tough

$$\vec{A} \times \vec{B} = \underbrace{AB \sin \theta}_{\text{magnitude}} \underbrace{\hat{n}}_{\text{Direction}}$$

Vector

Vector Product



② Dot product
Easy

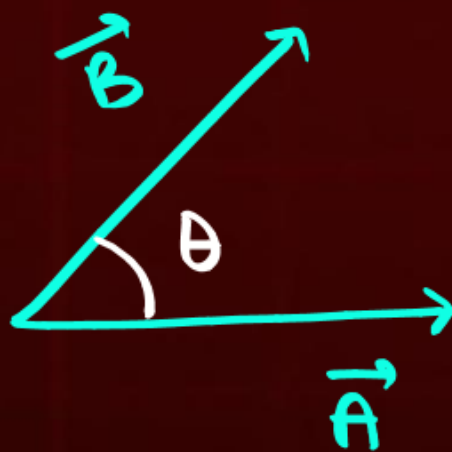
$$\vec{A} \cdot \vec{B} = \underbrace{AB \cos \theta}_m$$

Scalar product.

@ Cross product

/ vector product

①# The cross product of two vector is the multiplication of magnitude of two vector and Sin of Angle b/w them



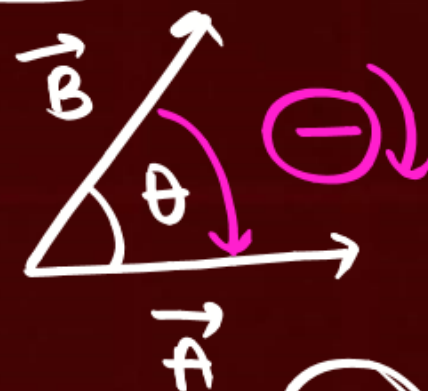
$$\underline{\vec{A} \times \vec{B}} = AB \sin \theta \hat{n}$$

(m) (D)

Vector product

②# The Direction of $\vec{A} \times \vec{B}$ is given by Screw Rule

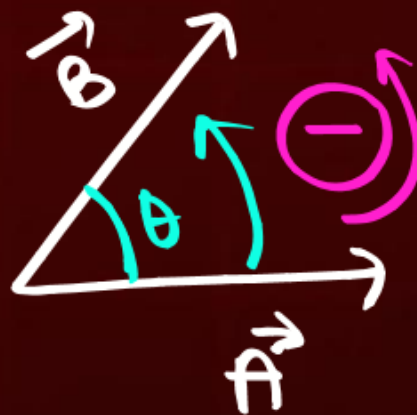
Case (i)



$$\vec{B} \times \vec{A} = BA \sin \theta \hat{n}$$

(D)
inward

Case (ii)



$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

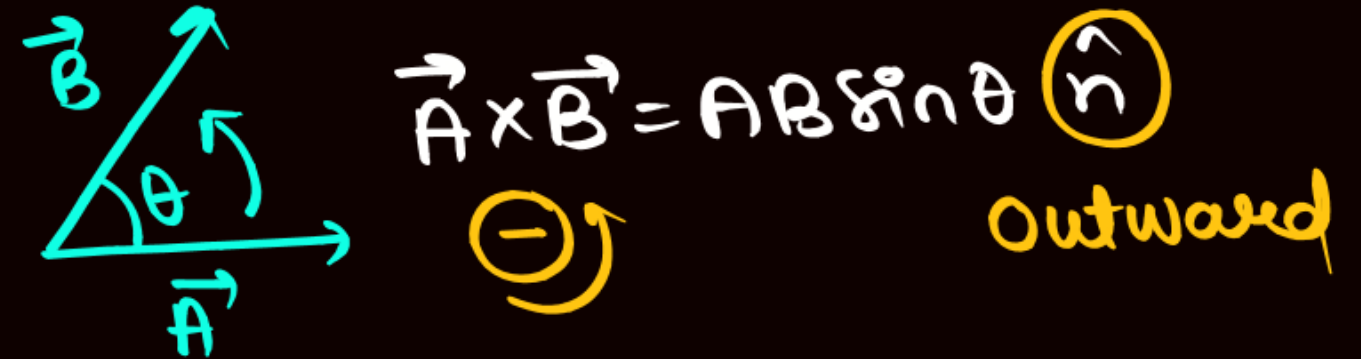
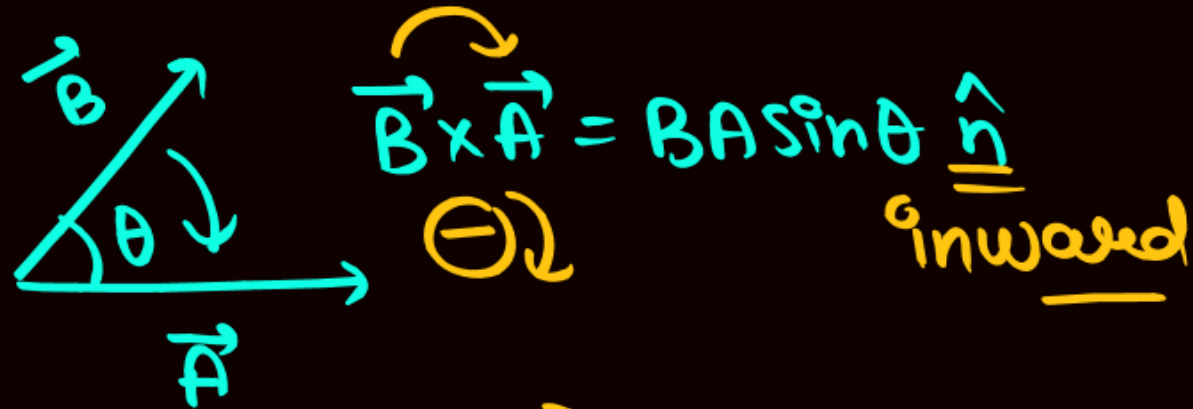
(D)
outward

Properties of Cross product

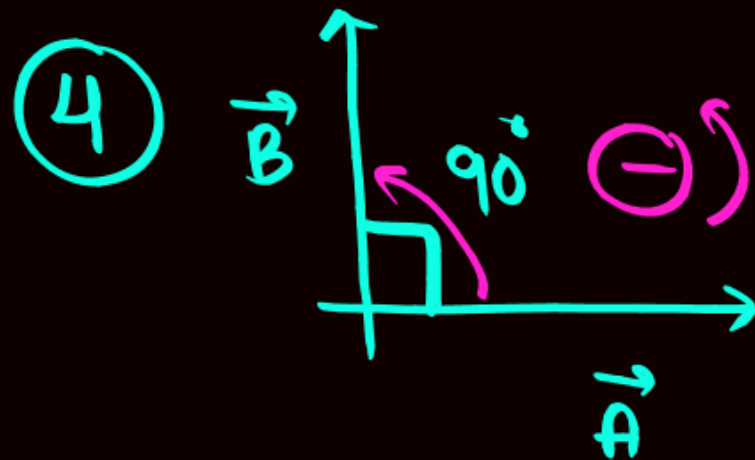
$$\frac{2 \times 4}{8} = \frac{4 \times 2}{8}$$

① $\vec{A} \times \vec{B} = \underline{AB \sin \theta} \hat{n}$ (Answer $\rightarrow$ Vector) It is the vector product.

② the direction of unit vector is given by Screw Rule

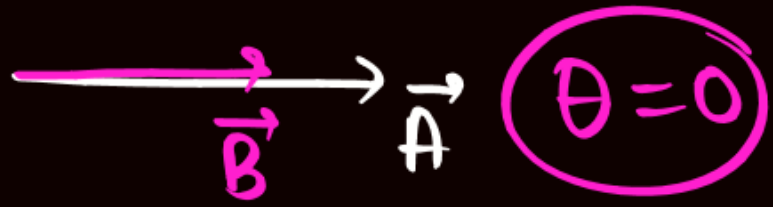


③ $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$ do not show commutative property.
out in



$$\begin{aligned} \vec{A} \times \vec{B} &= AB \sin \theta \hat{n} \\ &= AB \sin 90^\circ \hat{n} \\ &= AB \hat{n} \rightarrow \underline{\text{out}} \end{aligned}$$

⑤



$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

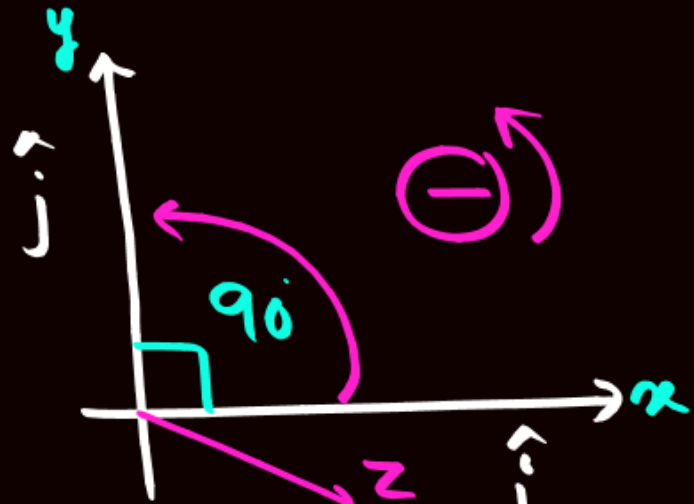
$$= AB \sin 0 \hat{n}$$

$$= AB \times 0 \times \hat{n}$$

$$= \underline{0}$$

Q① numerical (Neet-2017)

$$\hat{i} \times \hat{j} = \underline{\hspace{2cm}}$$



$$\begin{aligned}\hat{i} &= 1 \\ \hat{j} &= 1 \\ \theta &= 90^\circ\end{aligned}$$

Sol

$$\begin{aligned}\vec{A} \times \vec{B} &= AB \sin \theta \hat{n} \\ \underline{1\hat{i} \times 1\hat{j}} &= 1 \times 1 \sin 90^\circ \hat{n} \\ &= 1 \times 1 \times 1 \hat{n} \\ &= 1 \hat{n} \rightarrow \text{outward} \\ &= \underline{1\hat{k}} = \underline{\hat{k}}\end{aligned}$$

Q②

$$\hat{j} \times \hat{i} = \underline{\hspace{2cm}}$$



$$\begin{aligned}\hat{i} &= 1 \\ \hat{j} &= 1 \\ \theta &= 90^\circ\end{aligned}$$

$$\begin{aligned}\underline{\text{Sol}} \quad \vec{A} \times \vec{B} &= AB \sin \theta \hat{n} \\ 1\hat{j} \times 1\hat{i} &= 1 \times 1 \times \sin 90^\circ \hat{n} \\ &= 1 \times 1 \times 1 \hat{n} \\ &= 1 \hat{n} \rightarrow \text{inward} \\ &= 1 \times -\hat{k} \\ &= \underline{-\hat{k}} \checkmark\end{aligned}$$

Shaguftha Trick.

$$\checkmark \# \hat{i} \times \hat{j} = +\hat{k}$$

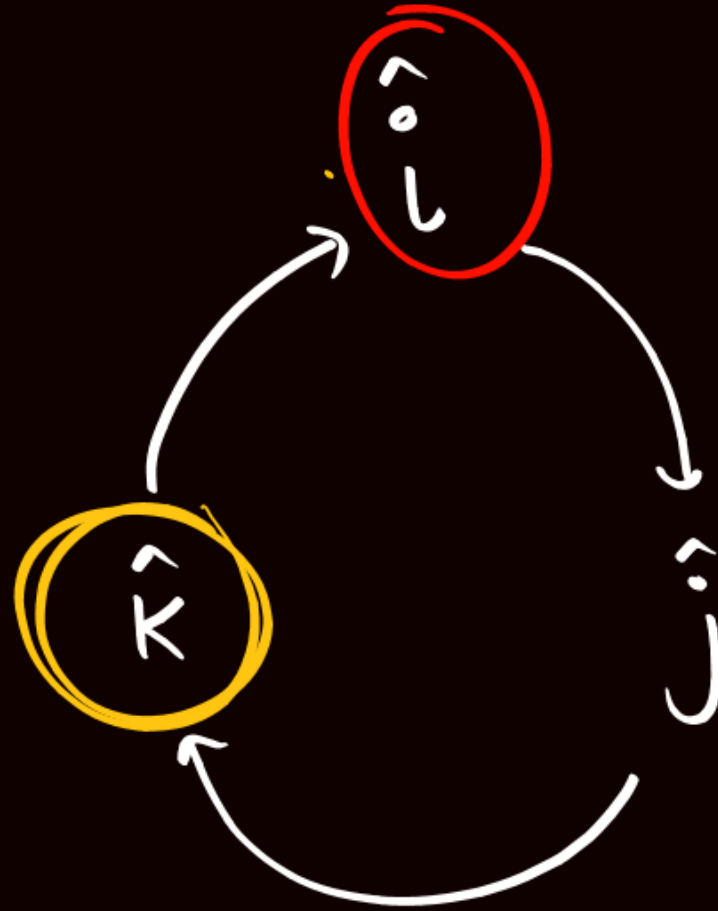
$$\checkmark \# \hat{j} \times \hat{i} = -\hat{k}$$

$$\text{u/w} \# \hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{j} = -\hat{i}$$

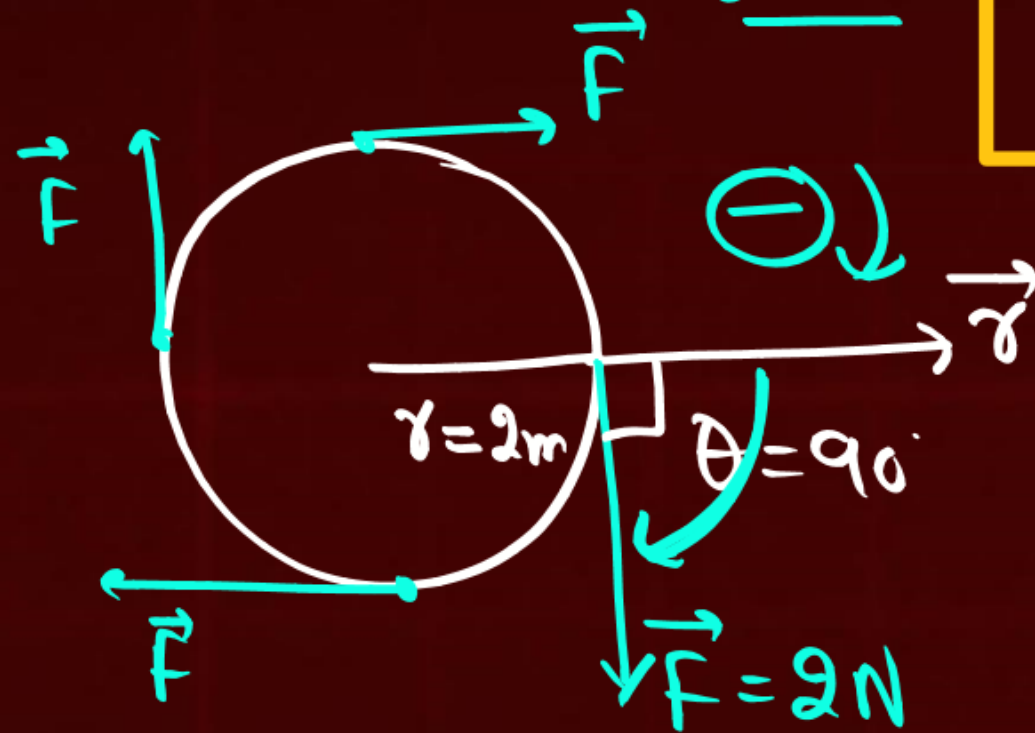
$$\# \hat{k} \times \hat{i} = \hat{j}$$

$$\hat{i} \times \hat{k} = -\hat{j}$$



Feel of cross product

Q numerical (Board) 2016



Calculate the Torque = ?

Rotation
effect
of force

Rotation

$$\# \text{ Torque} = \vec{r} \times \vec{F}$$

$$\tau = \vec{r} \times \vec{F}$$

$$\tau = r F \sin \theta \hat{n}$$

$$= 2 \times 2 \sin 90^\circ \hat{n}$$

$$= 2 \times 2 \times 1 \times \hat{n}$$

$$= 4 \hat{n} \rightarrow \text{inward}$$

$$= 4 (-\hat{k})$$

$$= \underline{\underline{-4 \hat{k}}}$$

H/W

When cross product is maximum?

$$\underline{\underline{\theta = 90^\circ}}$$

50/-

h/w

Blanks

501—

$$\hat{i} \times \hat{j} = \underline{\hspace{2cm}}$$

$$\hat{j} \times \hat{k} = \underline{\hspace{2cm}}$$

$$\hat{k} \times \hat{i} = \underline{\hspace{2cm}}$$

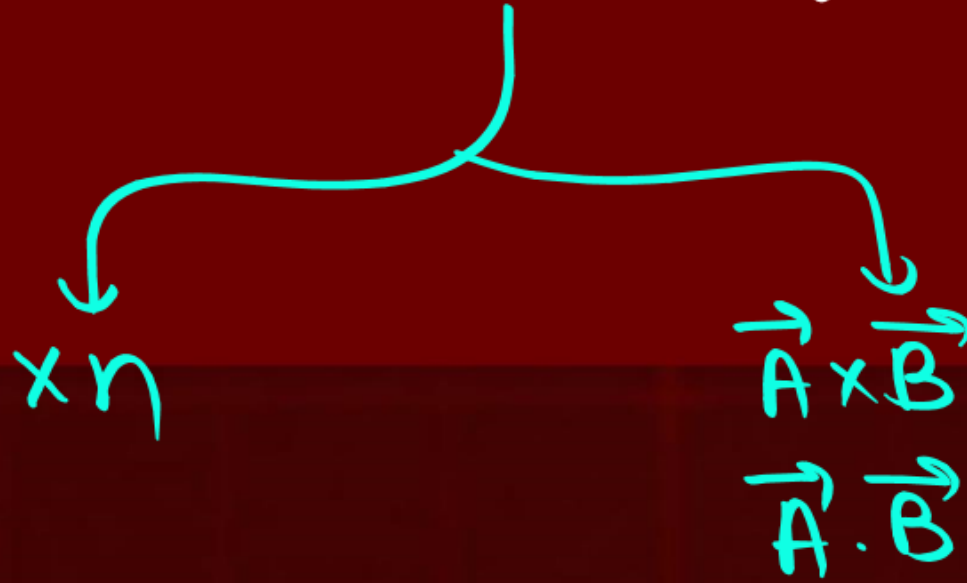
$$\hat{j} \times \hat{i} = \underline{\hspace{2cm}}$$

$$\hat{k} \times \hat{j} = \underline{\hspace{2cm}}$$

$$\hat{i} \times \hat{k} = \underline{\hspace{2cm}}$$

SUMMARY

multiplication of vector



Thank
You